Multiple Regression analysis

Housing market between 2001 and 2017 in Newcastle-Upon-Tyne

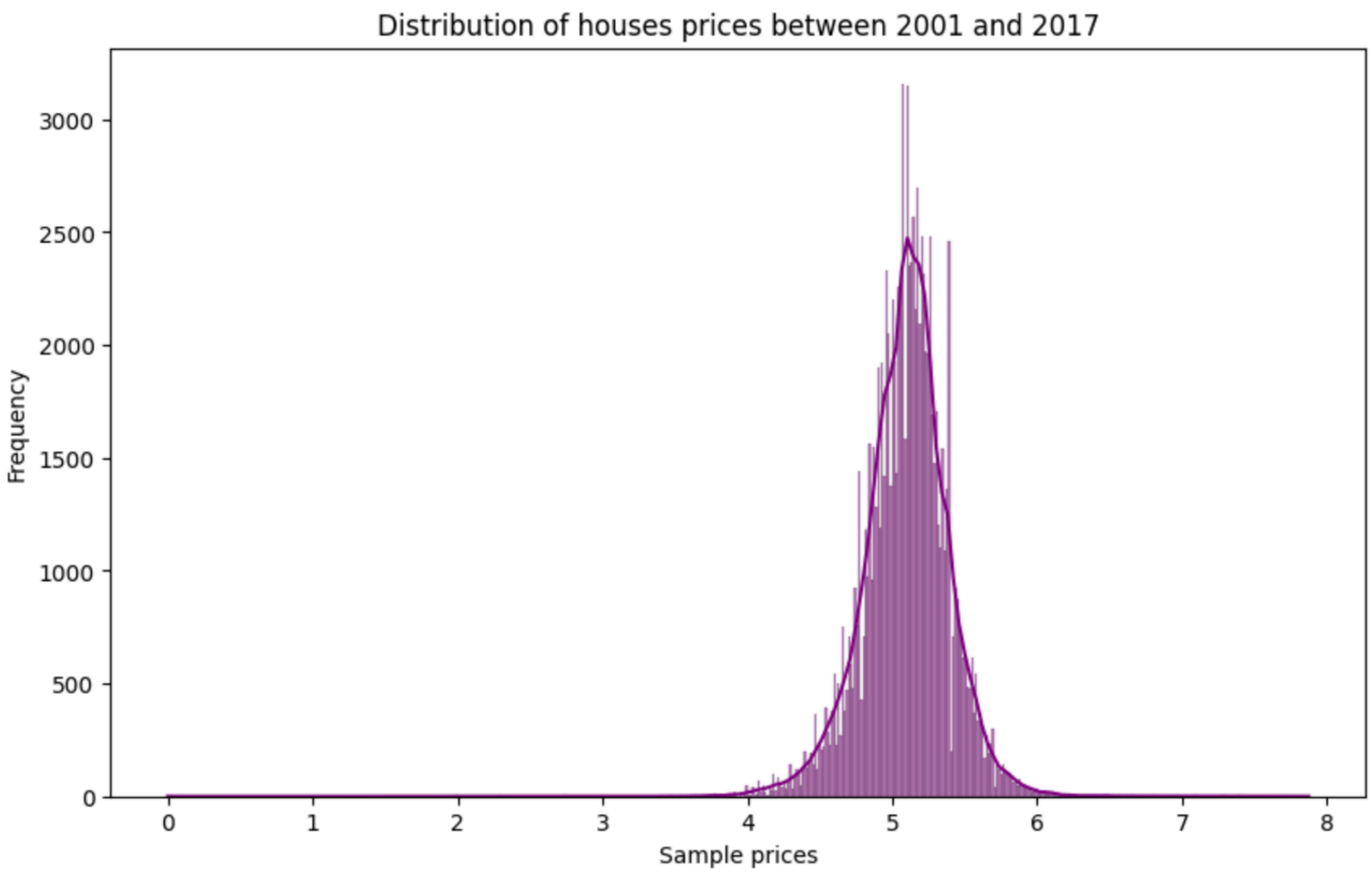
Dr Pat Ryser-Welch, July 2024

Let’s explore the data

What is your interpretation of this distribution, mean and standard deviation?

Newcastle-Upon-Type prices of properties exchange

Number of observations: 1.028080e+05
mean average (μ): 1.633868e+05
Standard deviation (σ) : 4.428592e+05



Interpretation

- The population size is reasonable with more than 10,000 independent observations. We can start to make inference about about the population from the sample.

- The shape of the distribution is unimodal, with a long tail on the left.

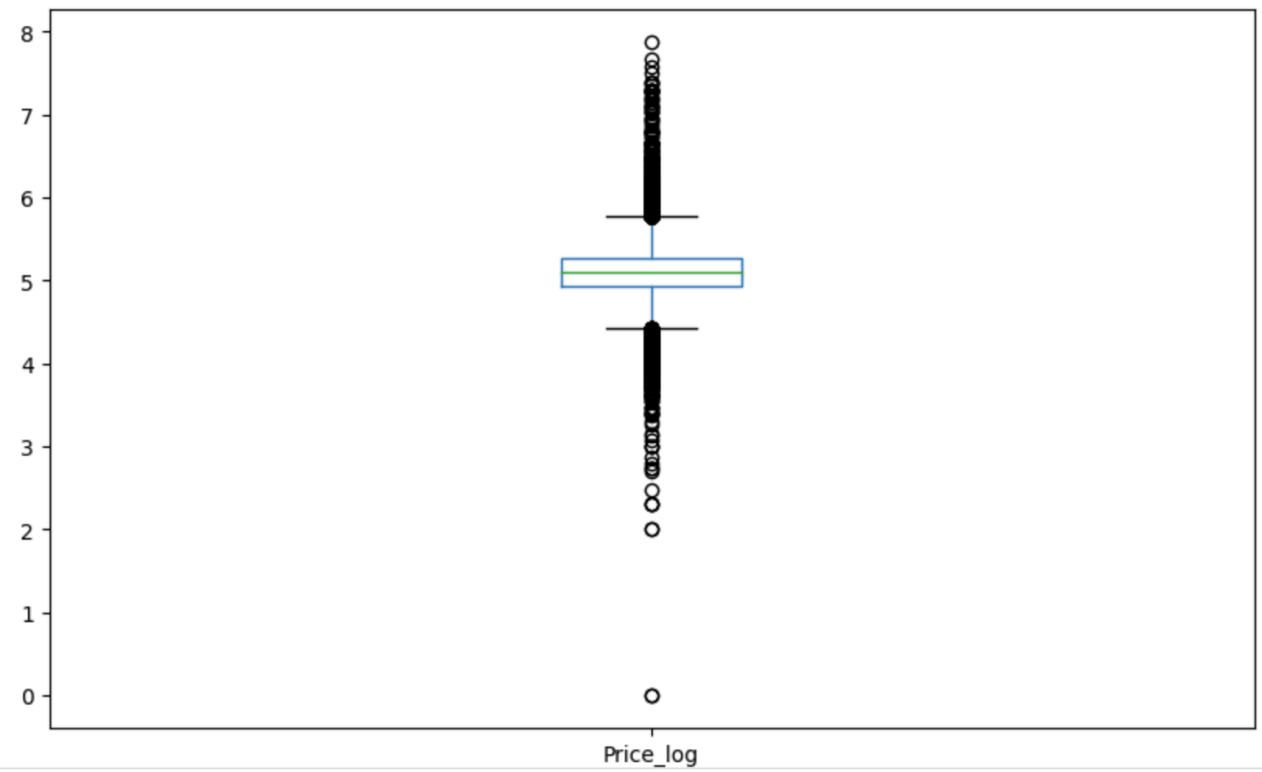
- The expected prices - population mean - is approximated to 160,000 (GBP).

- The large dispersion indicates variations of prices, which may be explained by type of properties and locations within Newcastle, some areas are cheaper than others.

What is your interpretation?

Non-parametric distribution of property prices in Newcastle

min	1.000000e+00
25%	8.500000e+04
50%	1.280000e+05
75%	1.850000e+05
max	7.493716e+07



What is your interpretation about the duration of ownership?

Duration	count	mean	min	25%	50%	75%	max	std
F	64736.0	180881.737565	1.0	90000.0	135997.5	209837.5	74937156.0	503344.192151
L	38071.0	133640.137427	100.0	78000.0	119500.0	162068.0	31600000.0	312109.141559
U	1.0	96000.000000	96000.0	96000.0	96000.0	96000.0	96000.0	NaN

Prices for newly built or not

Old/New	count	mean	min	25%	50%	75%	max	std
N	91449.0	160541.050968	1.0	82000.0	124950.0	180000.0	74937156.0	463969.977643
Y	11359.0	186297.320627	11000.0	122000.0	165000.0	224860.5	19200000.0	203508.998639

Prices per property type

Property Type	count	mean	min	25%	50%	75%	max	std
D	14034.0	2.806184e+05	1440.0	180000.0	240000.0	325758.75	4750000.0	1.853070e+05
F	27443.0	1.157368e+05	1.0	68950.0	106000.0	147500.00	1350000.0	6.793477e+04
O	904.0	1.205763e+06	100.0	94208.0	135290.0	448125.00	74937156.0	4.431054e+06
S	27158.0	1.498547e+05	2638.0	95000.0	132500.0	174000.00	1800165.0	9.930183e+04
T	33269.0	1.359627e+05	550.0	78000.0	116000.0	167500.00	1700000.0	9.328437e+04

Prices per month

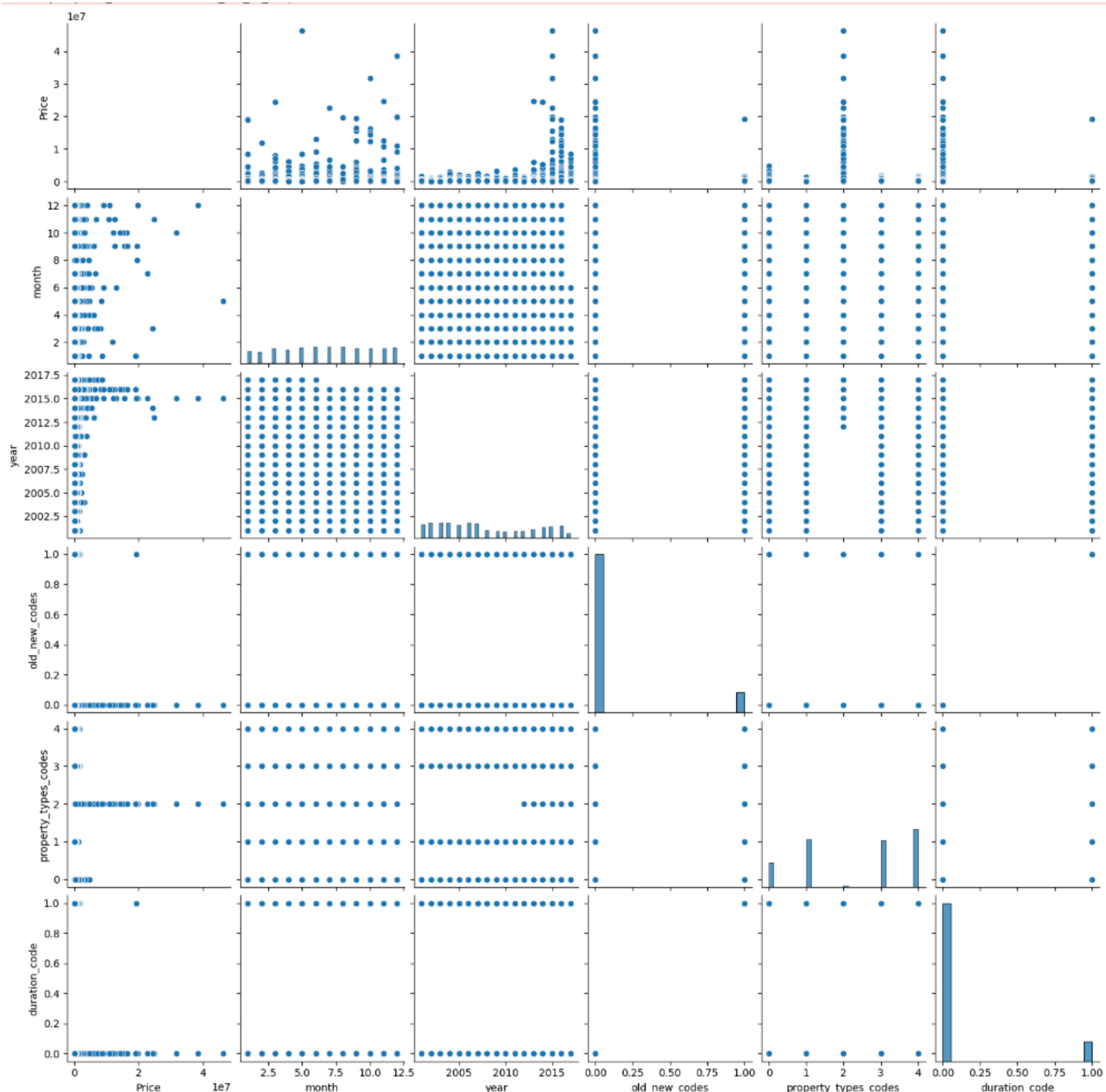
month	count	mean	min	25%	50%	75%	max	std
1	6733.0	173901.097134	1850.0	80000.0	124500.0	183000.0	74937156.0	1.009653e+06
2	6640.0	151060.400602	2700.0	80000.0	124688.5	177500.0	11862376.0	1.941405e+05
3	8707.0	160861.002067	750.0	81000.0	125000.0	185000.0	24300000.0	3.421922e+05
4	7969.0	156052.376082	630.0	83000.0	125000.0	182500.0	6150000.0	1.788920e+05
5	9004.0	160359.507552	1010.0	83375.0	127500.0	184950.0	46267545.0	5.137973e+05
6	9597.0	166134.440033	100.0	89950.0	132995.0	190000.0	13080000.0	2.325341e+05
7	9635.0	161264.782564	1.0	87000.0	130000.0	189995.0	22625000.0	2.749621e+05
8	9584.0	159549.801023	1000.0	89500.0	130000.0	188000.0	19580000.0	2.362472e+05
9	8682.0	165298.658028	1.0	87000.0	127800.0	184900.0	19500000.0	3.810666e+05
10	8755.0	166651.158424	500.0	83975.0	125000.0	185000.0	31600000.0	4.770786e+05
11	8572.0	165330.603826	1000.0	85000.0	127972.5	185000.0	24680000.0	4.556305e+05
12	8930.0	173214.566741	1178.0	90000.0	136795.0	194995.0	38479376.0	5.376368e+05

Prices per year

year	count	mean	min	25%	50%	75%	max	std
2001	7748.0	83066.740062	1.0	42475.0	67470.0	100000.0	1700000.0	6.592954e+04
2002	8517.0	99644.041916	3500.0	49950.0	80950.0	125000.0	1090000.0	7.715245e+04
2003	8495.0	124806.398117	1440.0	66500.0	103000.0	155000.0	1425000.0	9.258210e+04
2004	8359.0	148993.860629	10000.0	83050.0	129500.0	180000.0	3050000.0	1.094669e+05
2005	7360.0	154140.369701	10000.0	90000.0	132000.0	185000.0	2110000.0	1.079872e+05
2006	8666.0	165898.431341	13600.0	105000.0	143995.0	193000.0	1700000.0	1.045771e+05
2007	8102.0	175357.413972	10000.0	110000.0	145000.0	200000.0	2400000.0	1.195599e+05
2008	4164.0	179834.177233	20000.0	105000.0	145000.0	206000.0	1800000.0	1.371274e+05
2009	4031.0	168720.247085	1.0	100157.5	140000.0	195000.0	3186000.0	1.262644e+05
2010	3620.0	176722.460773	15000.0	100000.0	145000.0	213625.0	1200000.0	1.238542e+05
2011	3896.0	167182.490760	8050.0	95500.0	134000.0	197000.0	3750000.0	1.369242e+05
2012	3766.0	170396.999203	15000.0	100000.0	138000.0	200000.0	1420000.0	1.225406e+05
2013	4572.0	185840.270560	7400.0	100000.0	139950.0	202500.0	24680000.0	5.415061e+05
2014	6052.0	183869.773463	630.0	101000.0	145000.0	210000.0	24300000.0	3.643575e+05
2015	6308.0	244456.834337	2500.0	109000.0	153000.0	225000.0	46267545.0	1.121988e+06
2016	6733.0	241184.581613	100.0	106996.0	154500.0	225000.0	74937156.0	1.133514e+06
2017	2419.0	213530.547334	100.0	110000.0	153000.0	225000.0	8538000.0	3.818970e+05

To summarise

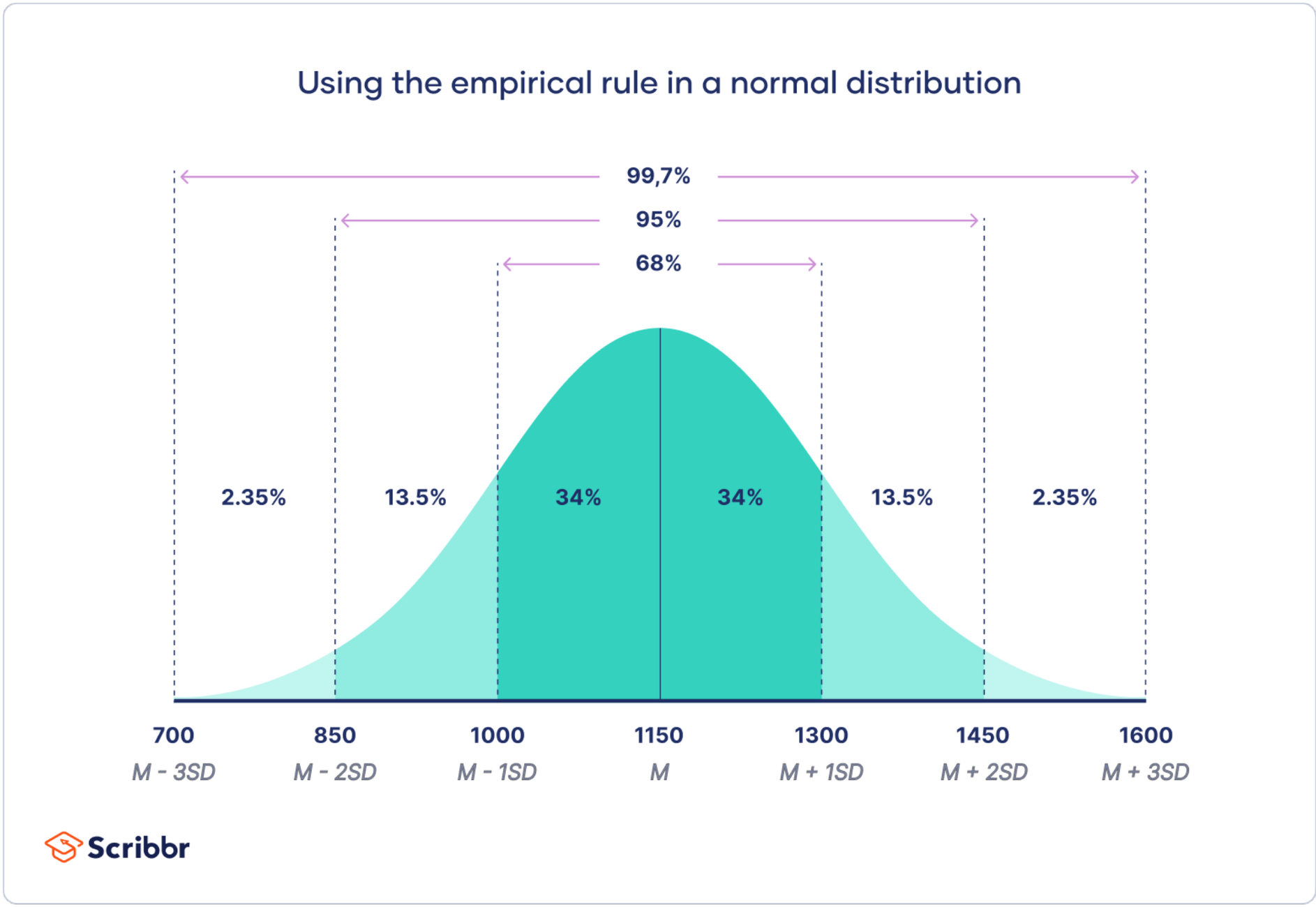
- There are many aspects an facets to data that may affect the prices.
- It appears some of those aspects may bring a higher level of complexity in a analysis.
- Between each statistical variables they appear to have repeated values.


1e7
Price
month
year
old_new_codes
property_types_codes
duration_code
Price
1e7
month
year
old_new_codes
property_types_codes
duration_code

Normal distribution, p-value, z-score

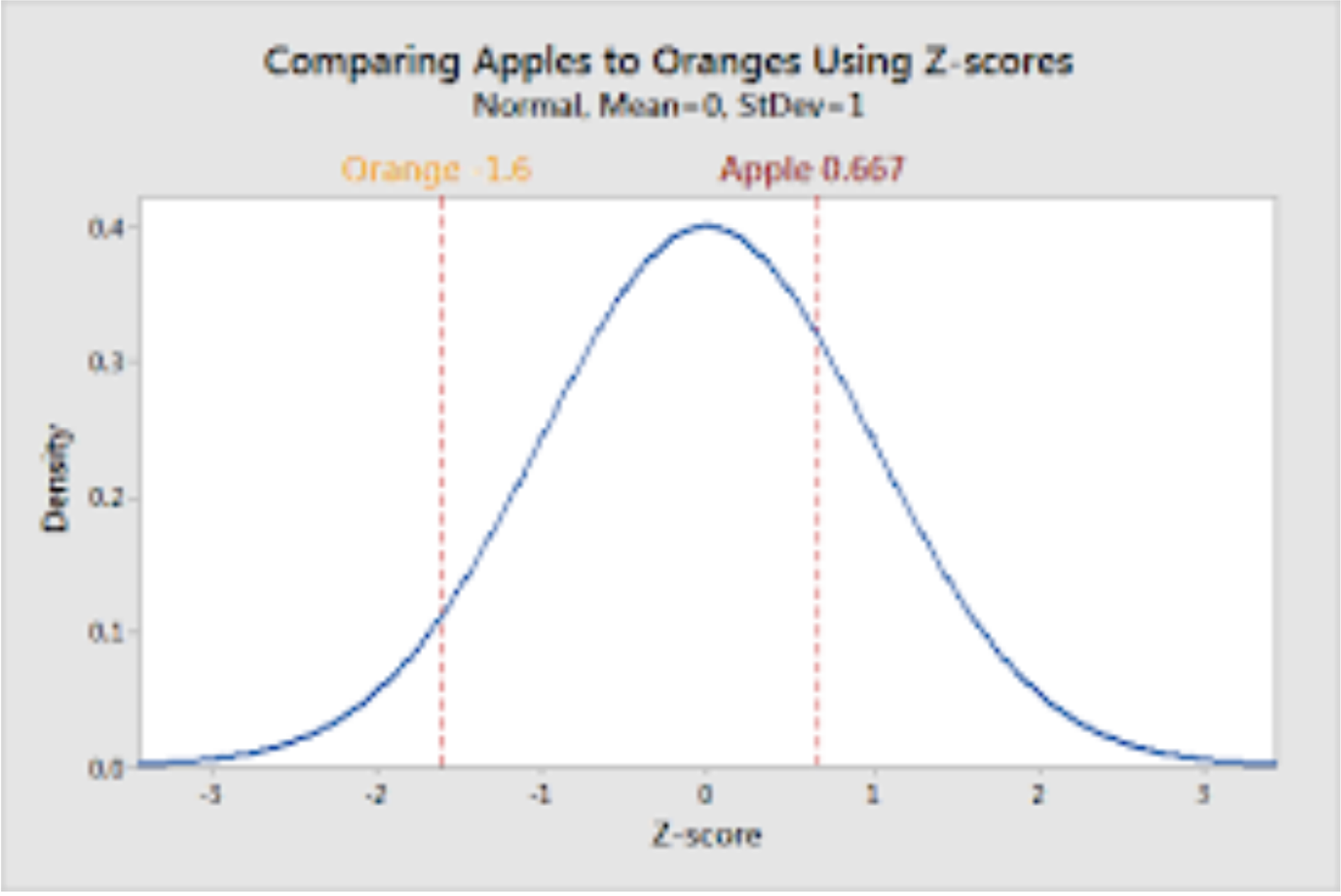
Normal distribution

A consolidation

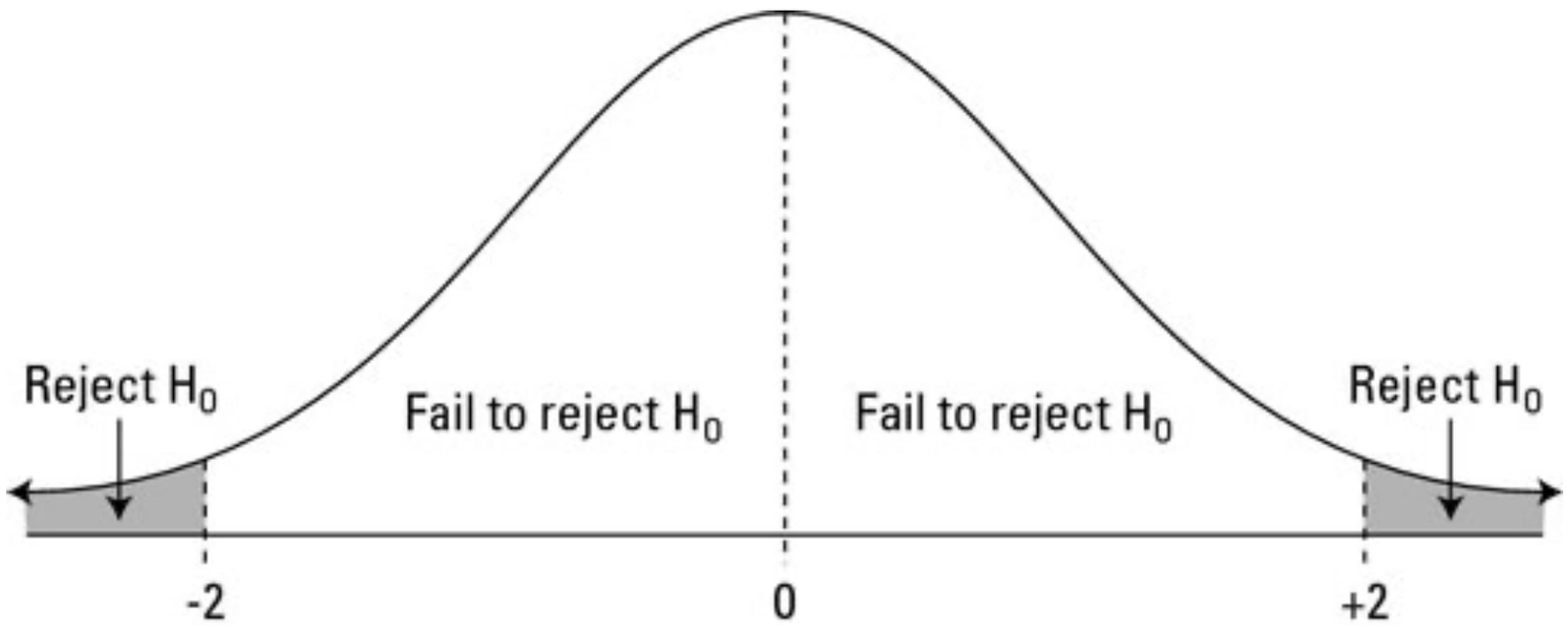


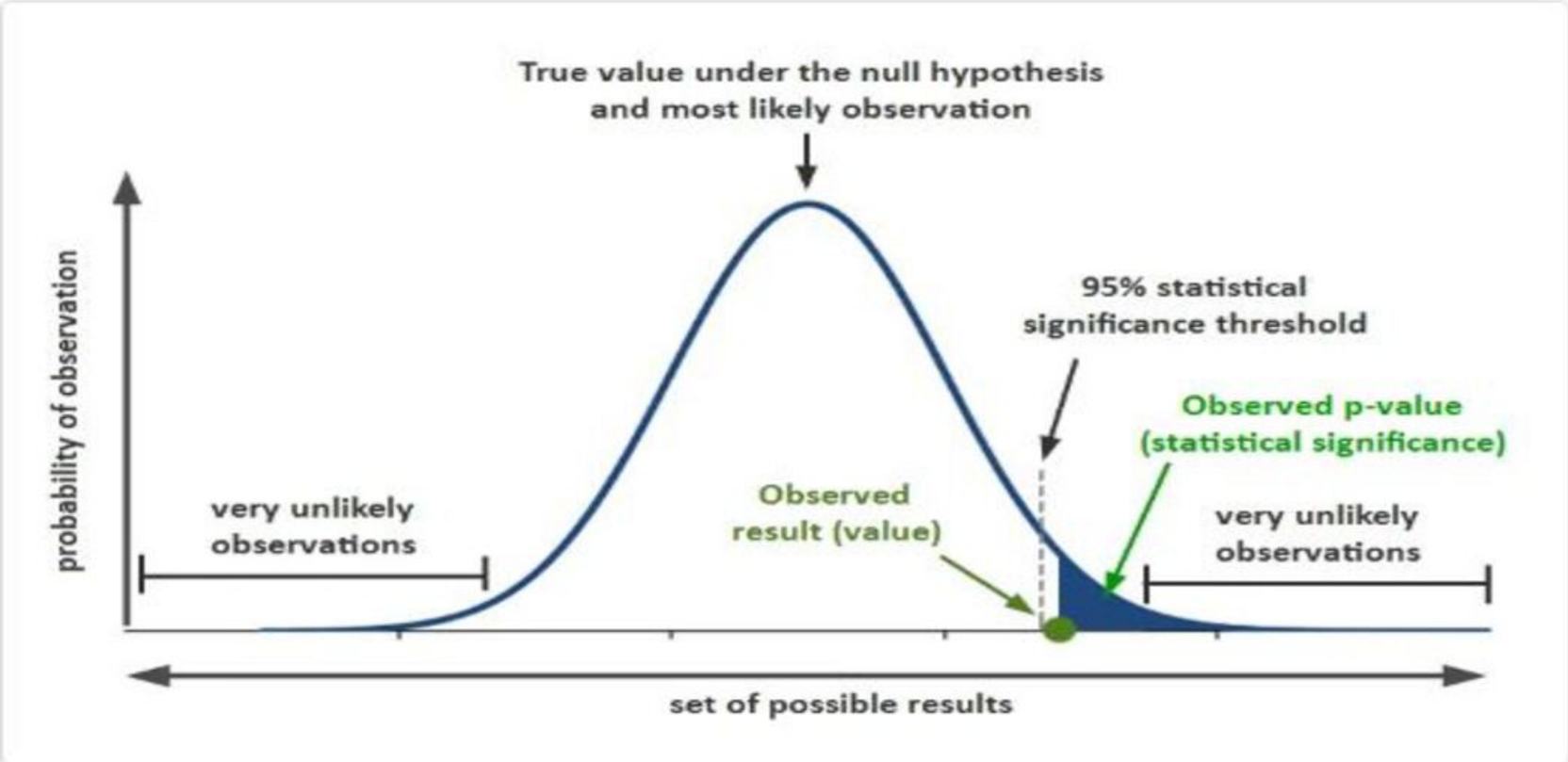
Z-score

Z-score = (data - mean) / std



Normal distributions and p-value





P-Value, and Hypothesis

A simple example

- Null means nothing:

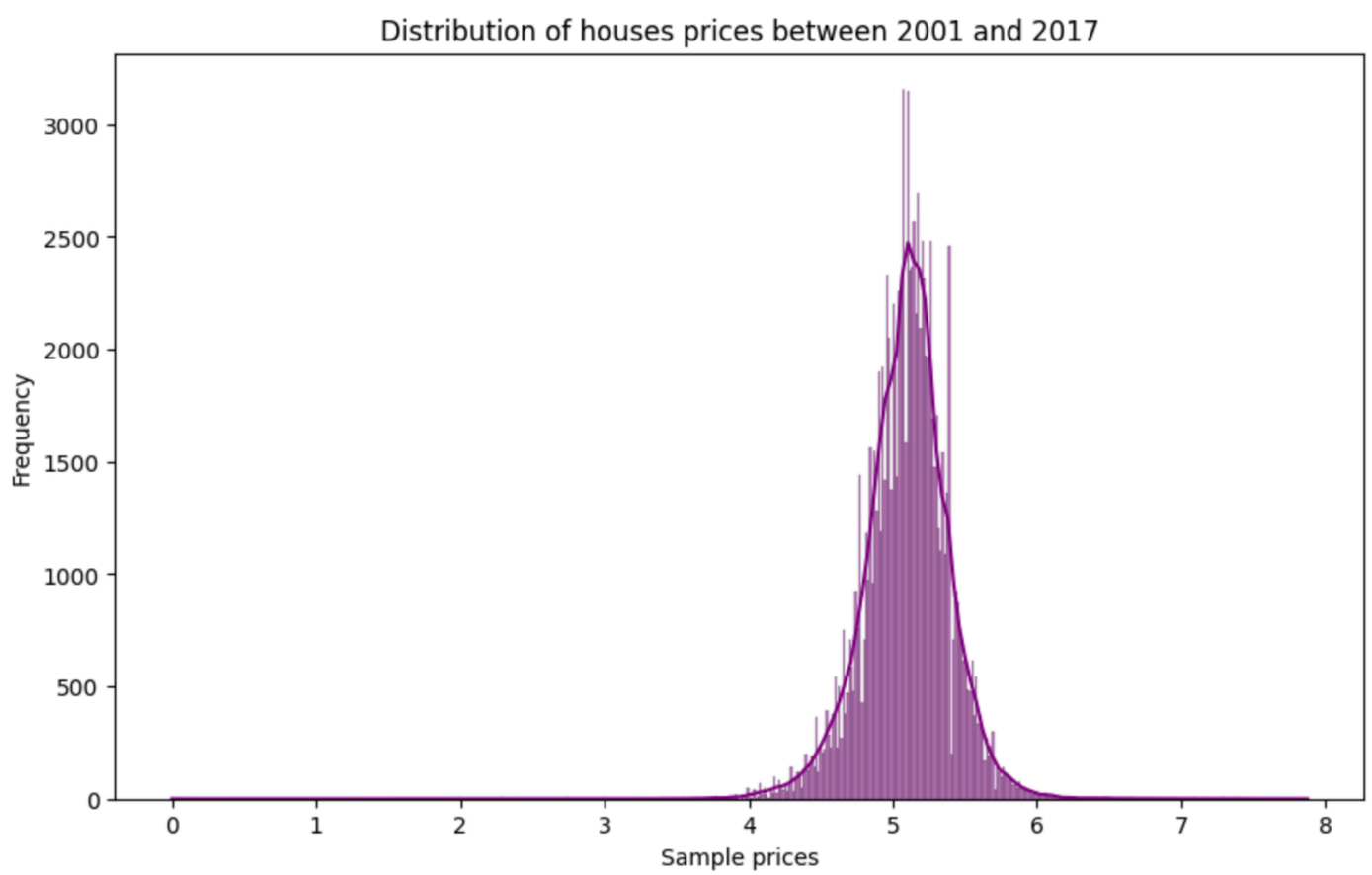
H_0: There is no difference in the salary of factory workers based on gender.

- Alternative known claim:

H_a: Male factory workers have a higher salary than female factory workers.

Two hypothesis

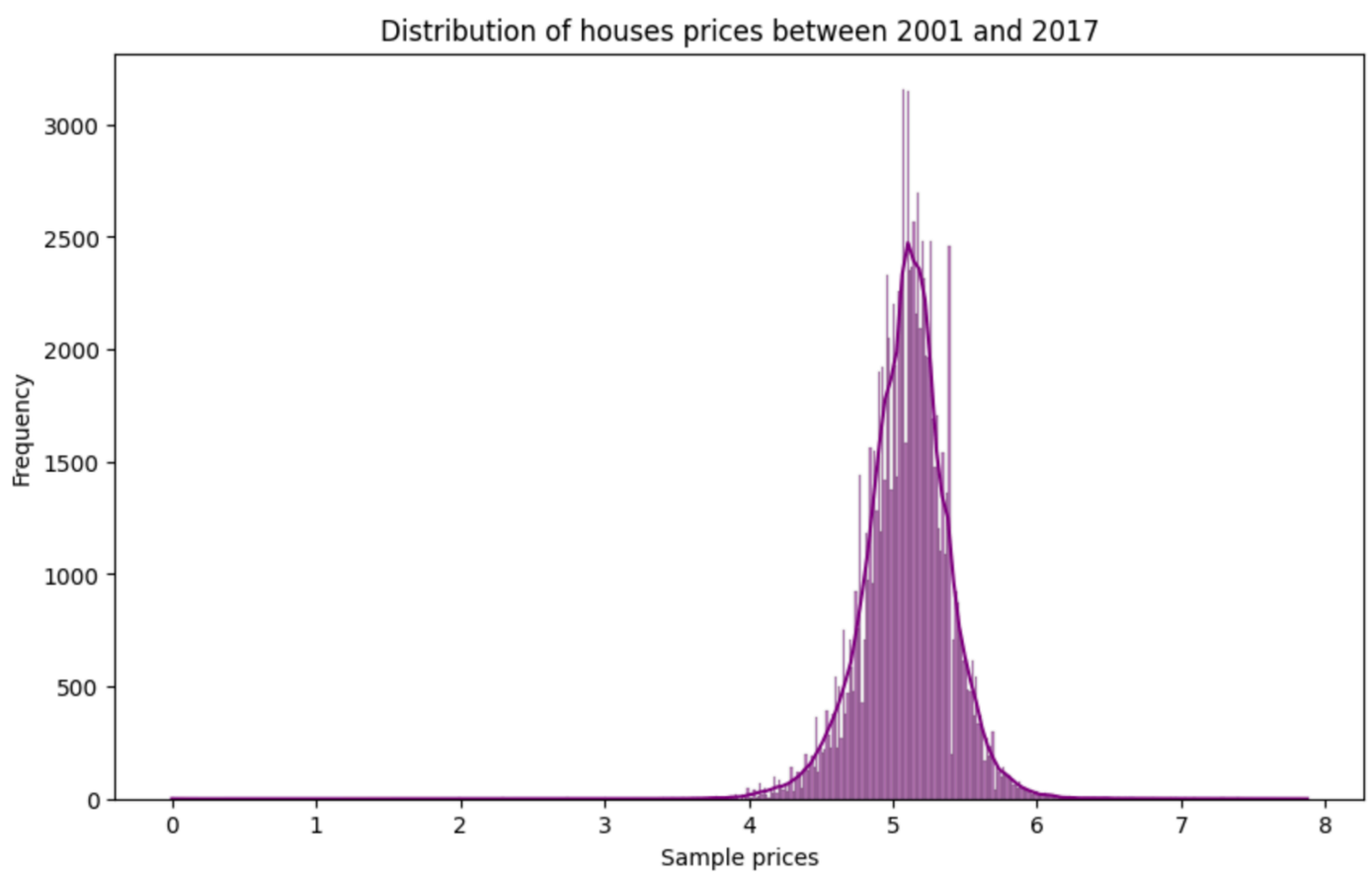
- Null hypothesis: People are prepared paying 1,500,000 (GDP) for a property in Newcastle
- Alternative hypothesis: People are not prepared paying 1,500,000 (GDP) for a property in Newcastle
- P-value = 0.05



Z-score
P = 0.002
P < 0.05
We reject the null hypothesis.
People are not prepared paying 1,500,000 (GDP)

Two hypothesis

- Null hypothesis: People are prepared paying 150,000 (GDP) for a property in Newcastle
- Alternative hypothesis: People are not prepared paying 150,000 (GDP) for a property in Newcastle
- P-value = 0.05



Z-score = 0.03
P = 0.98 (2. s.f)
P > 0.05
We <u>fail to reject</u> the null hypothesis. People are prepared paying 150,000 (GDP).

Is there a relationship …

A simple linear regression analysis

A simulation

$$E(Y|x) = \beta_0 + \beta_1 x$$

The structural model underlying a linear regression analysis is tha the explanatory and outcome variables are linearly related such tha the population mean of the outcome for any x value is $\beta_0 + \beta_1 x$.

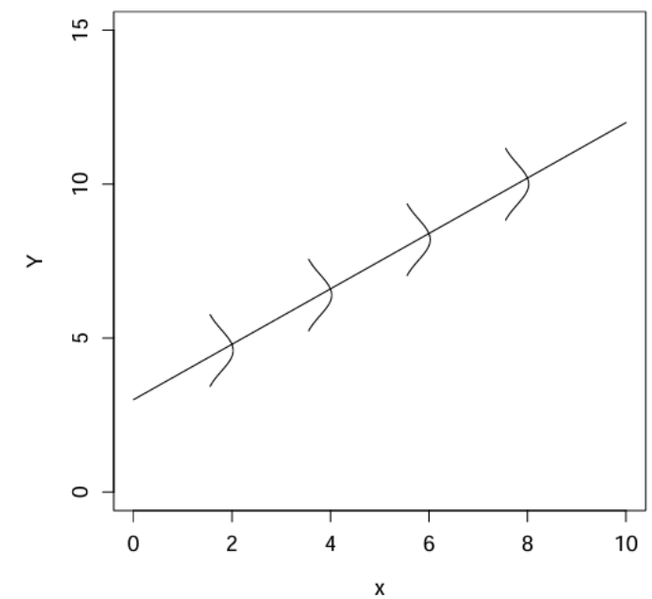
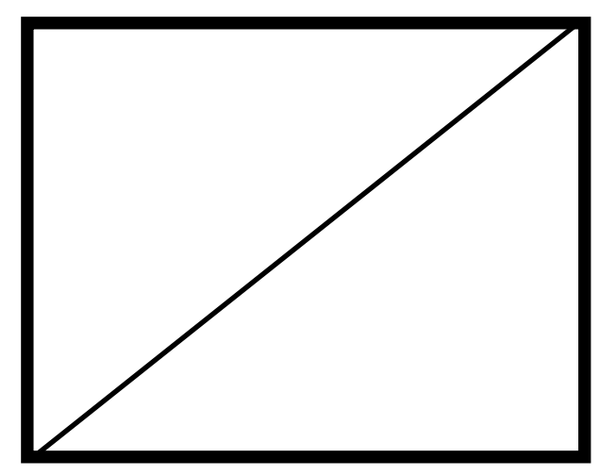


Figure 9.1: Mnemonic for the simple regression model.

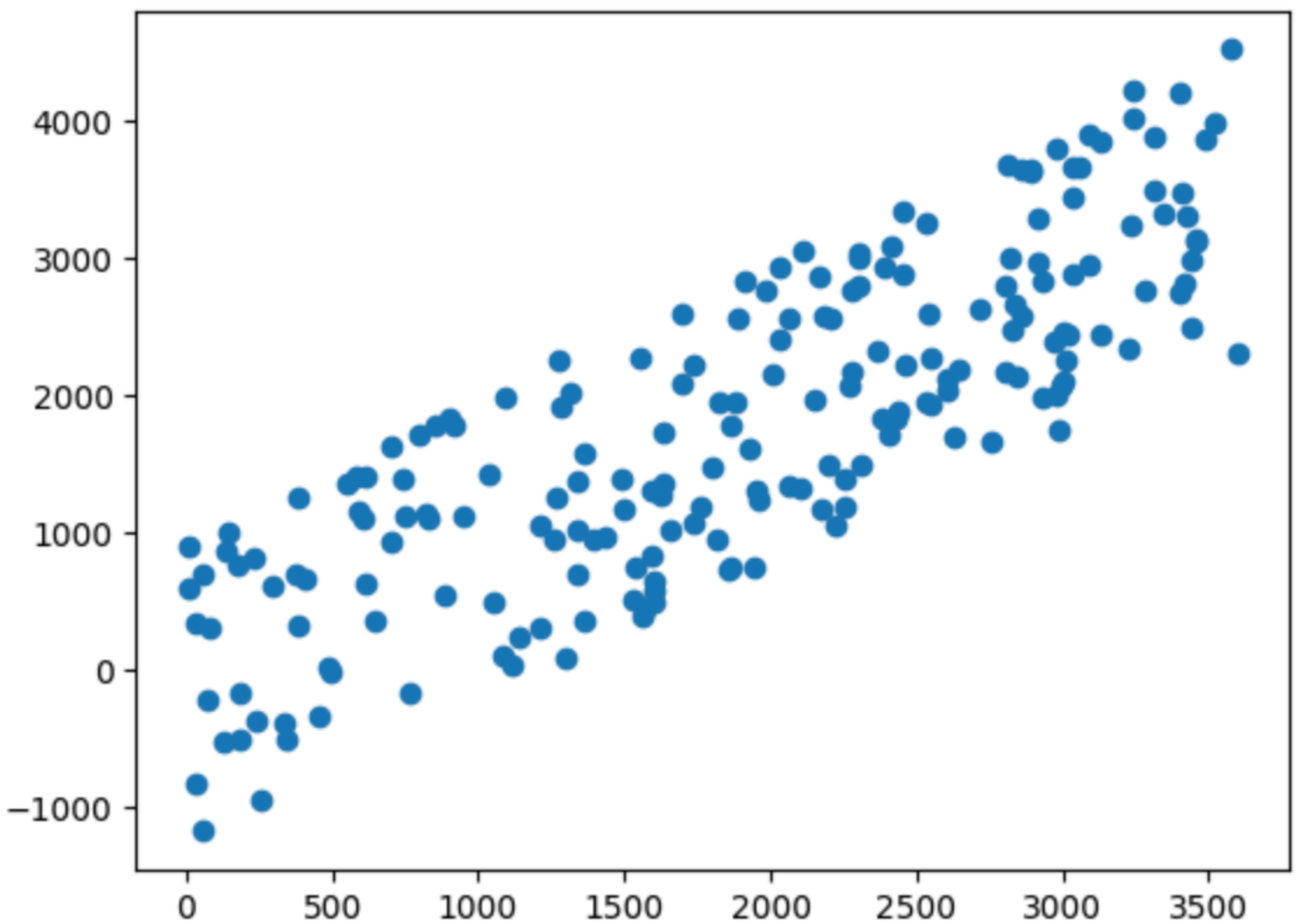
The usual regression null hypothesis is $H_0 : \beta_1 = 0$. Sometimes it is also meaningful to test $H_0 : \beta_0 = 0$ or $H_0 : \beta_1 = 1$.



https://www.stat.cmu.edu/~hseltman/309/Book/chapter9.pdf

Linear relationship or not?

Data generated randomly



Pearson : 0.85

Spearman : 0.85

Kendal : 0.65

Relationship ....

OLS Regression Results

Dep. Variable:	ys	**R-squared:**	0.710
Model:	OLS	**Adj. R-squared:**	0.709
Method:	Least Squares	**F-statistic:**	485.5
Date:	Sun, 28 Jul 2024	**Prob (F-statistic):**	3.63e-55
Time:	14:29:10	**Log-Likelihood:**	-1590.3
No. Observations:	200	**AIC:**	3185.
Df Residuals:	198	**BIC:**	3191.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	-248.5147	100.697	-2.468	0.014	-447.091	-49.939
xs	1.0714	0.049	22.033	0.000	0.976	1.167

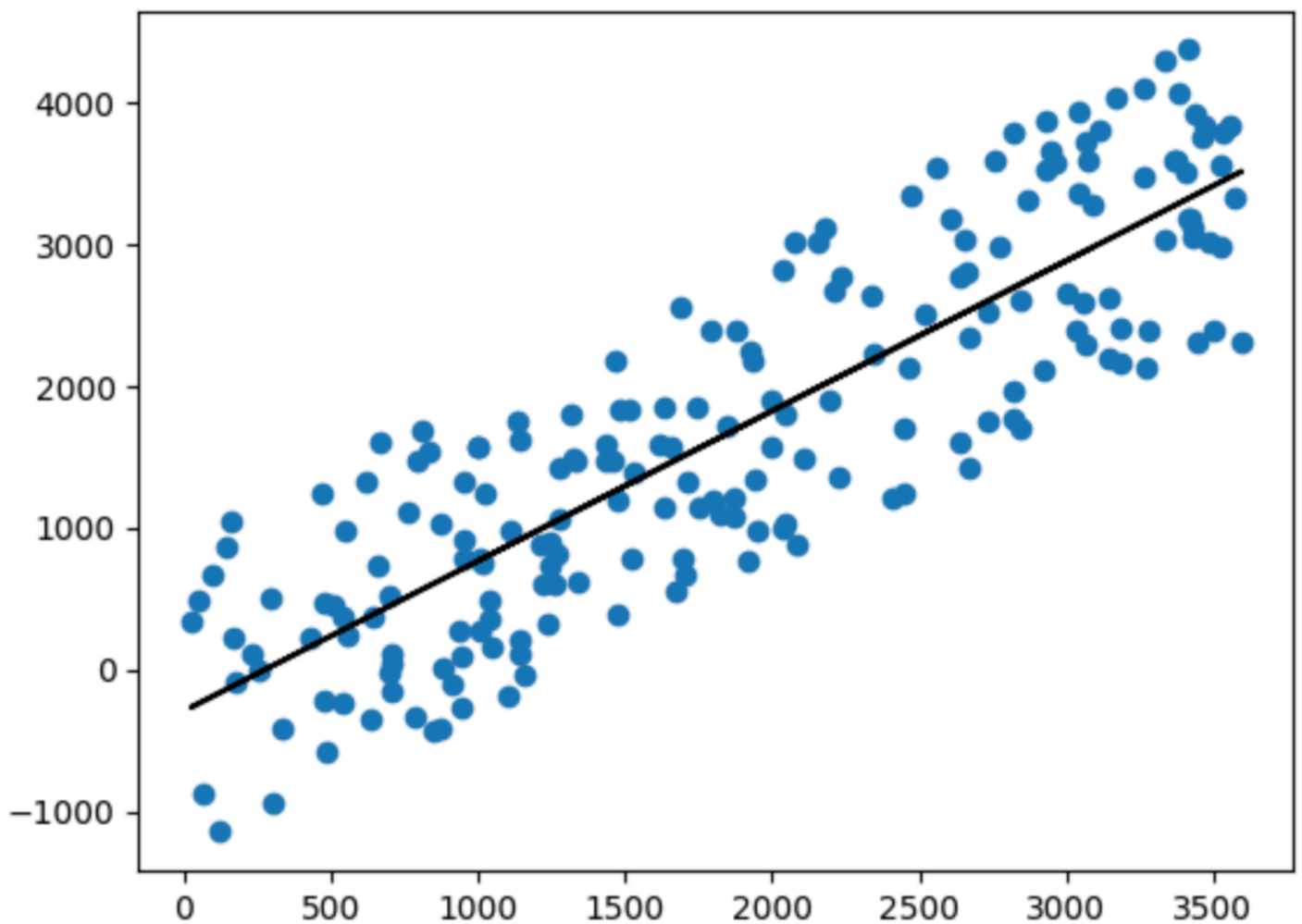
Omnibus:	101.769	**Durbin-Watson:**	1.966
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	12.396
Skew:	-0.041	**Prob(JB):**	0.00203
Kurtosis:	1.783	**Cond. No.**	4.27e+03

https://medium.com/swlh/interpreting-linear-regression-through-statsmodels-summary-4796d359035a

Let’s simulate a very good linear relationship

Data generated randomly

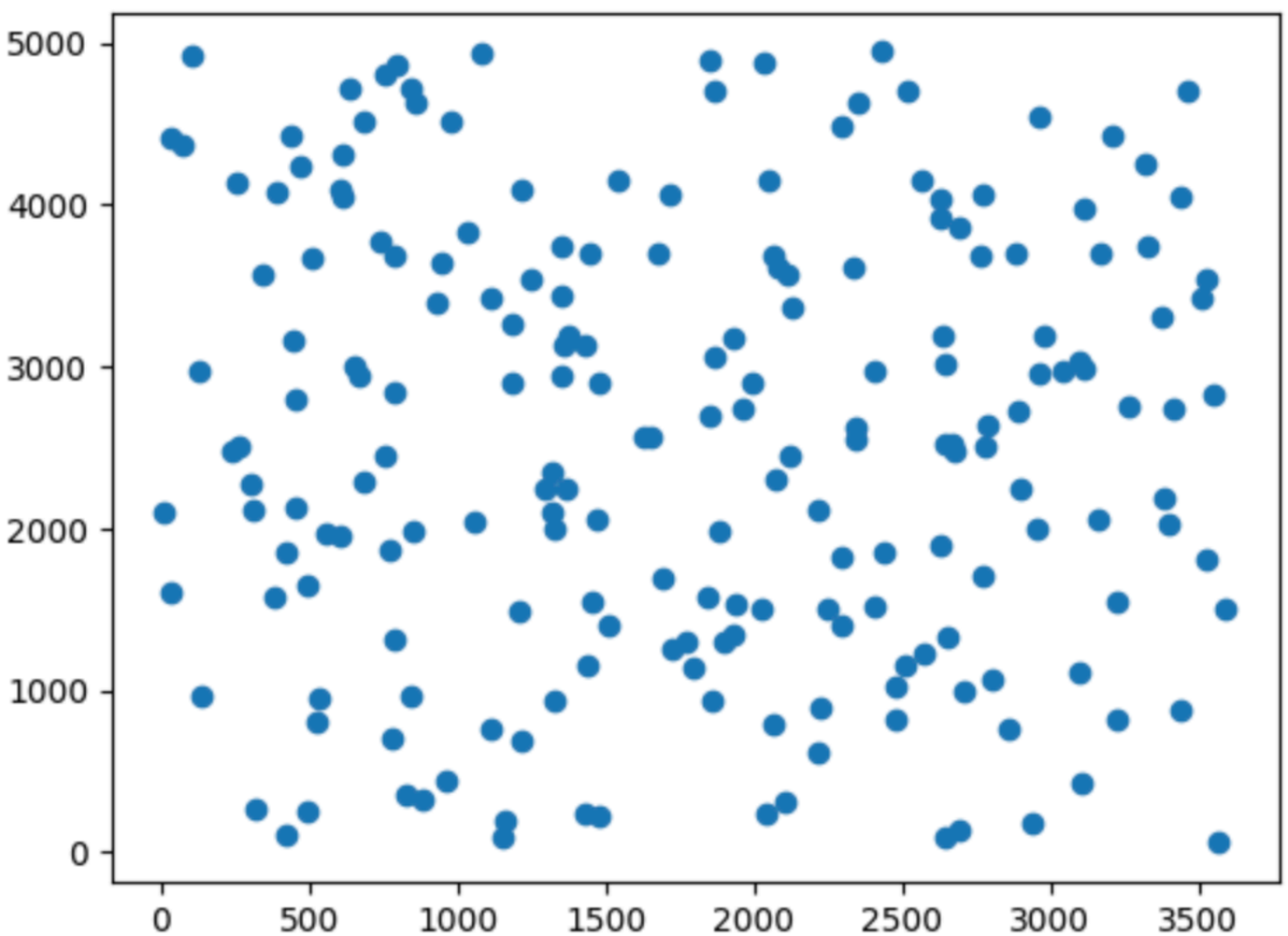
```
slope (B_1)     :  1.0575030603009183
intercept (B_0) :  -292.4051955859161
r               :  0.8588667747045551
p-value.        :  1.937492371369836e-59
standard error  :  0.044818983356903055
```



We can reject the null hypothesis $p < 0.05$. a predictor that has a low p-value is likely to be a meaningful addition to your model because changes in the predictor's value are related to changes in the response variable.

Linear relationship or not?

Data generated randomly



Pearson : 0.04

Spearman : 0.04

Kendal : 0.02

Relationship ....

OLS Regression Results

Dep. Variable:	ys	**R-squared:**	0.000
Model:	OLS	**Adj. R-squared:**	-0.005
Method:	Least Squares	**F-statistic:**	0.03677
Date:	Sun, 28 Jul 2024	**Prob (F-statistic):**	0.848
Time:	14:36:57	**Log-Likelihood:**	-1756.6
No. Observations:	200	**AIC:**	3517.
Df Residuals:	198	**BIC:**	3524.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	2405.8131	228.056	10.549	0.000	1956.082	2855.544
xs	0.0209	0.109	0.192	0.848	-0.194	0.236

Omnibus:	189.377	**Durbin-Watson:**	2.201
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	14.359
Skew:	-0.033	**Prob(JB):**	0.000762
Kurtosis:	1.689	**Cond. No.**	4.24e+03

https://medium.com/swlh/interpreting-linear-regression-through-statsmodels-summary-4796d359035a

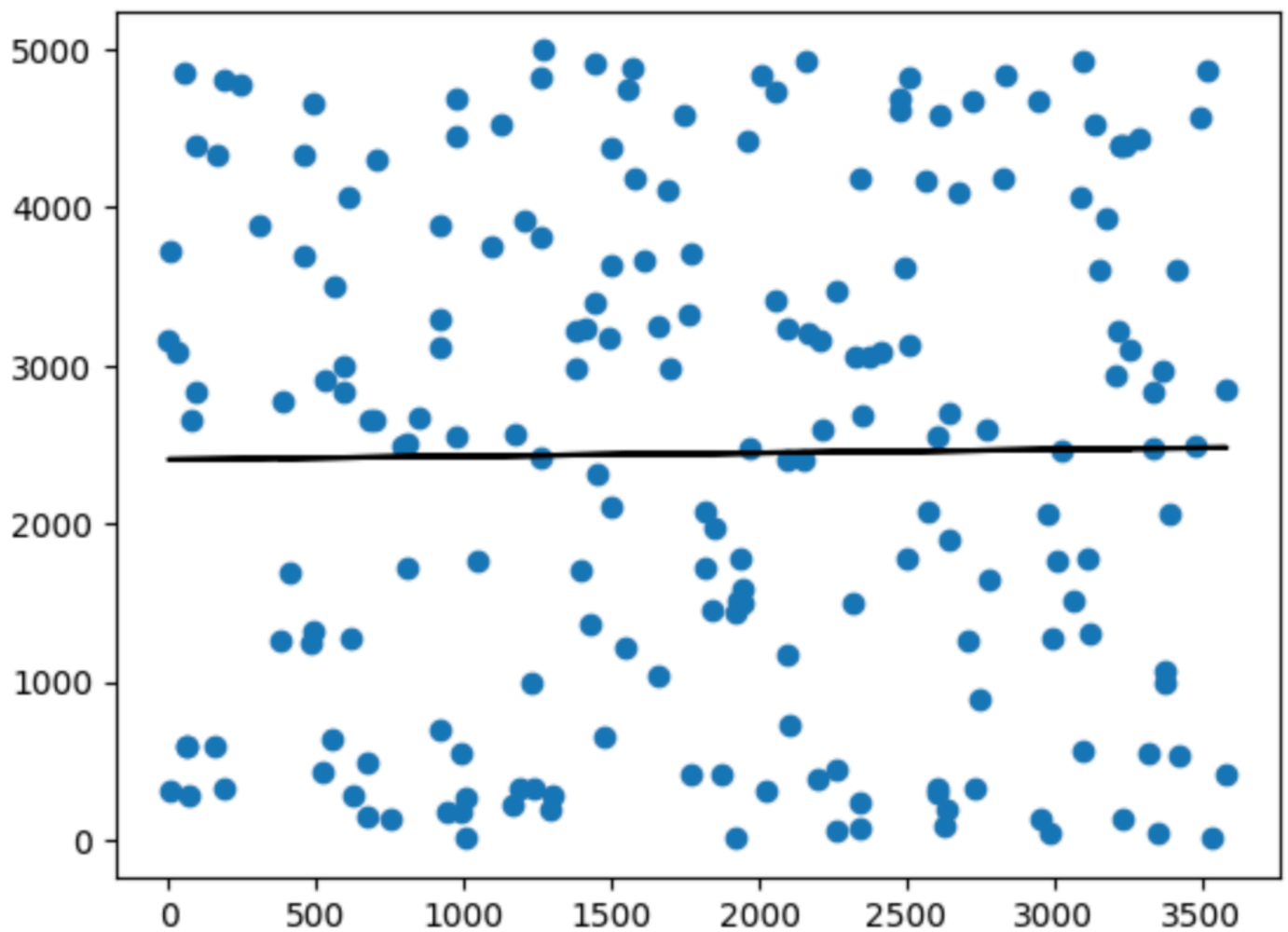
How to interpret?

- Look at the standard error
- Look at the p-value
- Look at the

Let’s simulate a very good linear relationship

Data generated randomly

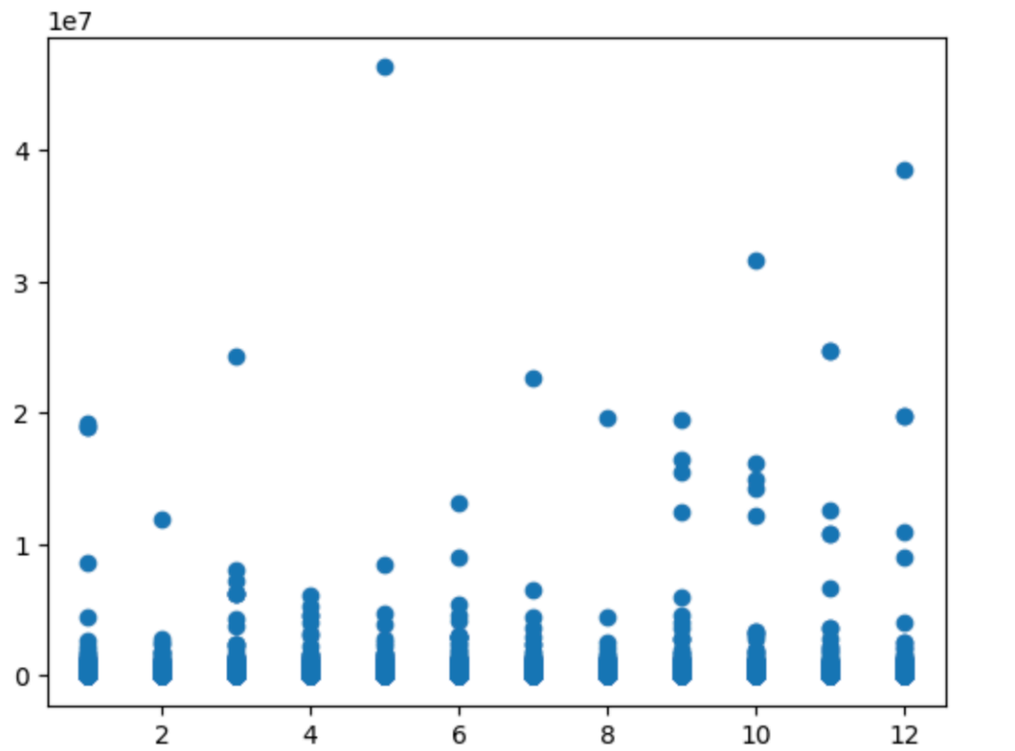
```
slope          :  0.020947038257367025
intercept      :  2405.813102266906
r              :  0.013627063435758877
p-value.       :  0.8481207822796989
standard error :  0.10923138135647471
```



We cannot reject the null hypothesis p > 0.05.

A predictor that has a high p-value is likely to not be a meaningful addition to your model. It does not change in the predictor's value are related to change in the response variable.

Is there a relationship between months and prices when purchasing a property in Newcastle?



OLS Regression Results

Dep. Variable:	Price	**R-squared:**	0.000
Model:	OLS	**Adj. R-squared:**	0.000
Method:	Least Squares	**F-statistic:**	10.32
Date:	Sun, 28 Jul 2024	**Prob (F-statistic):**	0.00131
Time:	14:58:21	**Log-Likelihood:**	-1.4658e+06
No. Observations:	102807	**AIC:**	2.932e+06
Df Residuals:	102805	**BIC:**	2.932e+06
Df Model:	1		
Covariance Type:	nonrobust		

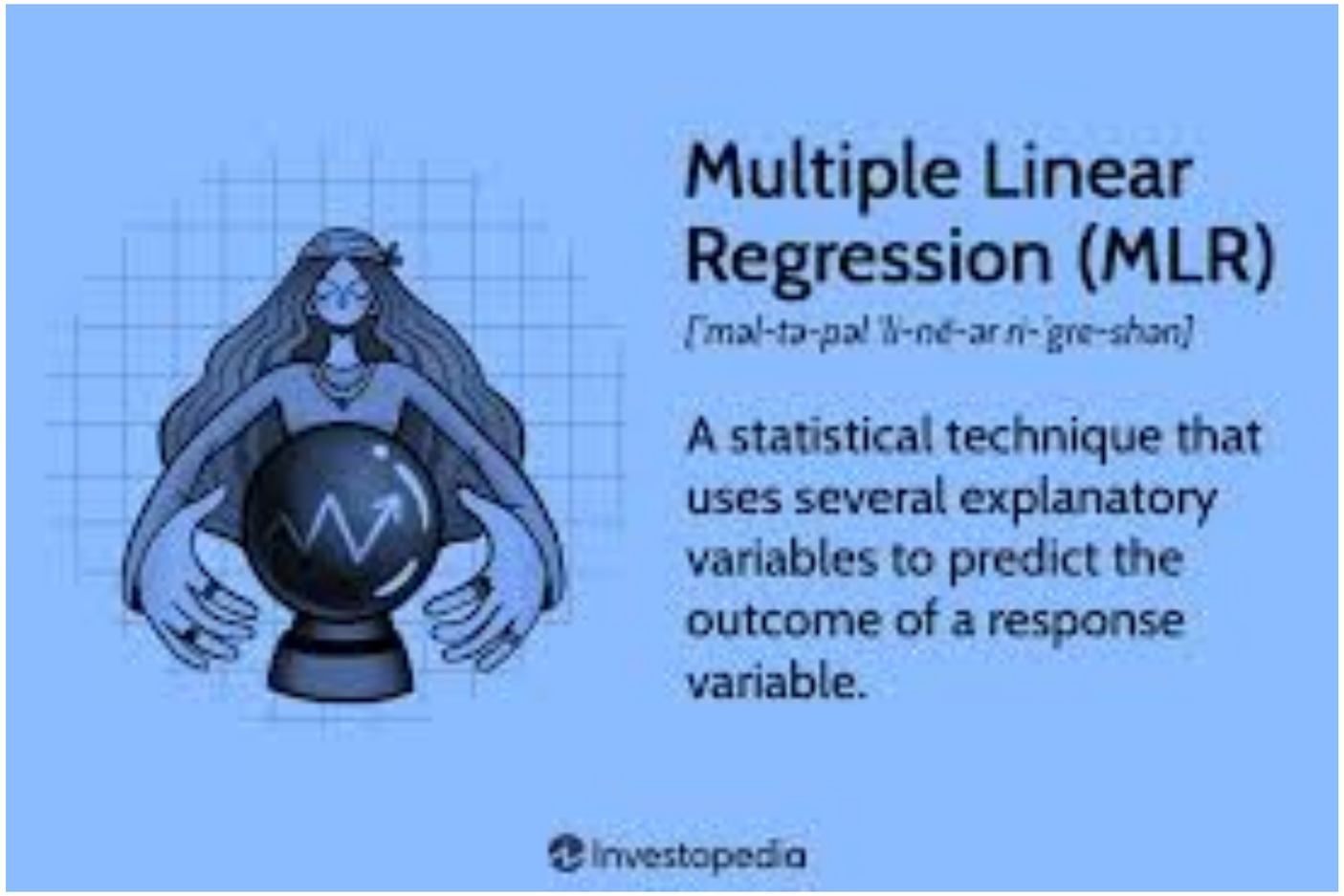
	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	1.55e+05	2646.544	58.582	0.000	1.5e+05	1.6e+05
month	1132.3372	352.439	3.213	0.001	441.560	1823.114

Omnibus:	344637.142	**Durbin-Watson:**	1.900
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	108685197163.037
Skew:	60.008	**Prob(JB):**	0.00
Kurtosis:	5038.657	**Cond. No.**	17.2

```
slope          :  -0.06401718361705189
intercept      :  2008.3161889831895
r              :  -0.043059968340976355
p-value.       :  2.1341967010983083e-43
standard error :  0.0046324713201987095
```

We can reject the null hypothesis $p < 0.05$, with some caution.

Multiple regression



Multiple regression analysis: Used to estimate the relationship between a dependent variable and two or more independent variables;

https://www.investopedia.com/terms/m/mlr.asp

Is there a relationship between months, years, types of properties, and duration and prices when purchasing a property in Newcastle?

OLS Regression Results

Dep. Variable:	Price	**R-squared:**	0.016
Model:	OLS	**Adj. R-squared:**	0.016
Method:	Least Squares	**F-statistic:**	410.9
Date:	Sun, 28 Jul 2024	**Prob (F-statistic):**	0.00
Time:	15:29:10	**Log-Likelihood:**	-1.4650e+06
No. Observations:	102807	**AIC:**	2.930e+06
Df Residuals:	102802	**BIC:**	2.930e+06
Df Model:	4		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	-1.541e+07	4.73e+05	-32.566	0.000	-1.63e+07	-1.45e+07
month	1588.8373	350.238	4.536	0.000	902.375	2275.300
year	7772.4667	235.668	32.981	0.000	7310.560	8234.374
property_types_codes	-1.774e+04	792.123	-22.398	0.000	-1.93e+04	-1.62e+04
duration_code	5547.1349	3782.371	1.467	0.142	-1866.263	1.3e+04

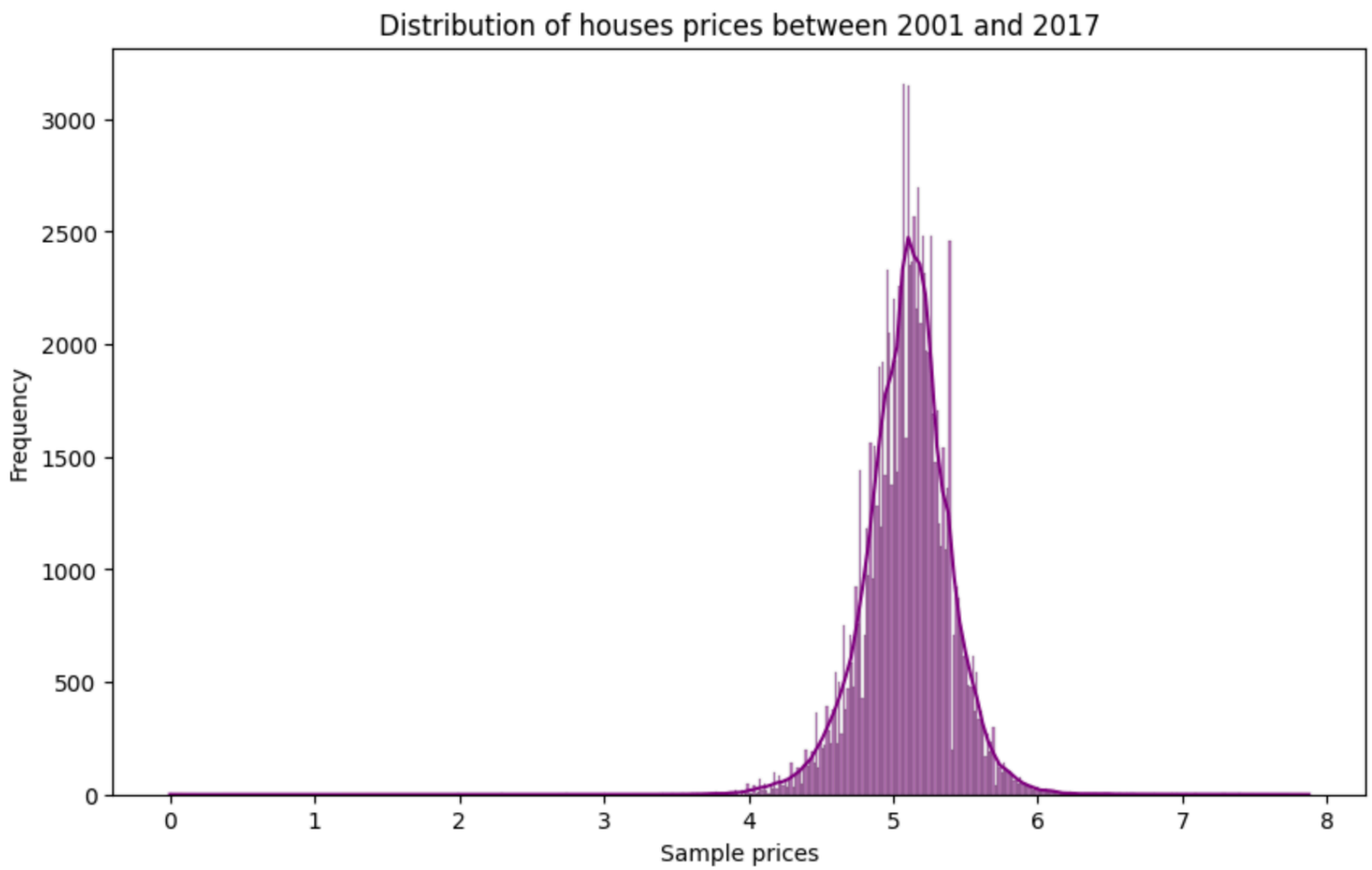
Omnibus:	346519.543	**Durbin-Watson:**	1.920
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	113981741804.934
Skew:	60.971	**Prob(JB):**	0.00
Kurtosis:	5159.922	**Cond. No.**	8.16e+05

$$E(Price) = -1.54e7 + .1.588373e3(Month) + 7.7724667e3(Year) + -1.774e4(types) + 5.5471349(Duration)$$

Between 2001 and 2017

Distribution

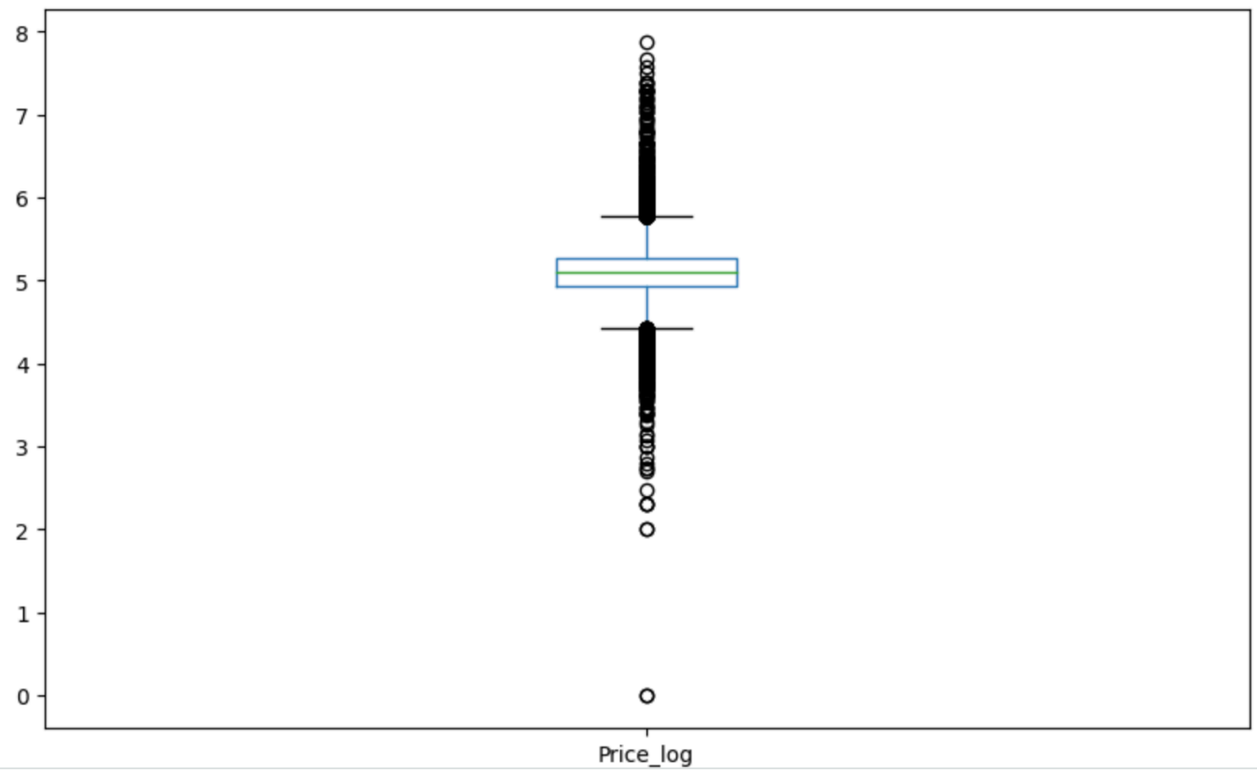
Number of observations: 1.028080e+05
mean average (μ): 1.633868e+05
Standard deviation (σ) : 4.428592e+05



Between 2001 and 2017

Non-parametric distribution

min	1.000000e+00
25%	8.500000e+04
50%	1.280000e+05
75%	1.850000e+05
max	7.493716e+07



Prices per Duration

Duration	count	mean	min	25%	50%	75%	max	std
F	64736.0	180881.737565	1.0	90000.0	135997.5	209837.5	74937156.0	503344.192151
L	38071.0	133640.137427	100.0	78000.0	119500.0	162068.0	31600000.0	312109.141559
U	1.0	96000.000000	96000.0	96000.0	96000.0	96000.0	96000.0	NaN

Prices for newly built or not

Old/New	count	mean	min	25%	50%	75%	max	std
N	91449.0	160541.050968	1.0	82000.0	124950.0	180000.0	74937156.0	463969.977643
Y	11359.0	186297.320627	11000.0	122000.0	165000.0	224860.5	19200000.0	203508.998639

Prices per property type

Property Type	count	mean	min	25%	50%	75%	max	std
D	14034.0	2.806184e+05	1440.0	180000.0	240000.0	325758.75	4750000.0	1.853070e+05
F	27443.0	1.157368e+05	1.0	68950.0	106000.0	147500.00	1350000.0	6.793477e+04
O	904.0	1.205763e+06	100.0	94208.0	135290.0	448125.00	74937156.0	4.431054e+06
S	27158.0	1.498547e+05	2638.0	95000.0	132500.0	174000.00	1800165.0	9.930183e+04
T	33269.0	1.359627e+05	550.0	78000.0	116000.0	167500.00	1700000.0	9.328437e+04

Prices per month

month	count	mean	min	25%	50%	75%	max	std
1	6733.0	173901.097134	1850.0	80000.0	124500.0	183000.0	74937156.0	1.009653e+06
2	6640.0	151060.400602	2700.0	80000.0	124688.5	177500.0	11862376.0	1.941405e+05
3	8707.0	160861.002067	750.0	81000.0	125000.0	185000.0	24300000.0	3.421922e+05
4	7969.0	156052.376082	630.0	83000.0	125000.0	182500.0	6150000.0	1.788920e+05
5	9004.0	160359.507552	1010.0	83375.0	127500.0	184950.0	46267545.0	5.137973e+05
6	9597.0	166134.440033	100.0	89950.0	132995.0	190000.0	13080000.0	2.325341e+05
7	9635.0	161264.782564	1.0	87000.0	130000.0	189995.0	22625000.0	2.749621e+05
8	9584.0	159549.801023	1000.0	89500.0	130000.0	188000.0	19580000.0	2.362472e+05
9	8682.0	165298.658028	1.0	87000.0	127800.0	184900.0	19500000.0	3.810666e+05
10	8755.0	166651.158424	500.0	83975.0	125000.0	185000.0	31600000.0	4.770786e+05
11	8572.0	165330.603826	1000.0	85000.0	127972.5	185000.0	24680000.0	4.556305e+05
12	8930.0	173214.566741	1178.0	90000.0	136795.0	194995.0	38479376.0	5.376368e+05

Prices per year

year	count	mean	min	25%	50%	75%	max	std
2001	7748.0	83066.740062	1.0	42475.0	67470.0	100000.0	1700000.0	6.592954e+04
2002	8517.0	99644.041916	3500.0	49950.0	80950.0	125000.0	1090000.0	7.715245e+04
2003	8495.0	124806.398117	1440.0	66500.0	103000.0	155000.0	1425000.0	9.258210e+04
2004	8359.0	148993.860629	10000.0	83050.0	129500.0	180000.0	3050000.0	1.094669e+05
2005	7360.0	154140.369701	10000.0	90000.0	132000.0	185000.0	2110000.0	1.079872e+05
2006	8666.0	165898.431341	13600.0	105000.0	143995.0	193000.0	1700000.0	1.045771e+05
2007	8102.0	175357.413972	10000.0	110000.0	145000.0	200000.0	2400000.0	1.195599e+05
2008	4164.0	179834.177233	20000.0	105000.0	145000.0	206000.0	1800000.0	1.371274e+05
2009	4031.0	168720.247085	1.0	100157.5	140000.0	195000.0	3186000.0	1.262644e+05
2010	3620.0	176722.460773	15000.0	100000.0	145000.0	213625.0	1200000.0	1.238542e+05
2011	3896.0	167182.490760	8050.0	95500.0	134000.0	197000.0	3750000.0	1.369242e+05
2012	3766.0	170396.999203	15000.0	100000.0	138000.0	200000.0	1420000.0	1.225406e+05
2013	4572.0	185840.270560	7400.0	100000.0	139950.0	202500.0	24680000.0	5.415061e+05
2014	6052.0	183869.773463	630.0	101000.0	145000.0	210000.0	24300000.0	3.643575e+05
2015	6308.0	244456.834337	2500.0	109000.0	153000.0	225000.0	46267545.0	1.121988e+06
2016	6733.0	241184.581613	100.0	106996.0	154500.0	225000.0	74937156.0	1.133514e+06
2017	2419.0	213530.547334	100.0	110000.0	153000.0	225000.0	8538000.0	3.818970e+05